Manufacturing & Video-Analytics Intelligent Report

“I want the data from the mes application for machine utilization and from video analytics I want camera configuration information”

Generated: 2026-10-17 17:49:40
Time filter applied

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Executive Summary

2	12	333	Applied	0
DATABASES QUERIED	TABLES MATCHED	TOTAL RECORDS	TIME FILTER	FAILED QUERIES

Matched keywords: camera, configuration, machine, utilization

Telemetry window: 2026-10-16T12:19:40Z → 2026-10-17T12:19:40Z (bucketed every 5m)

Video Analytics Camera Configuration and Status Logs

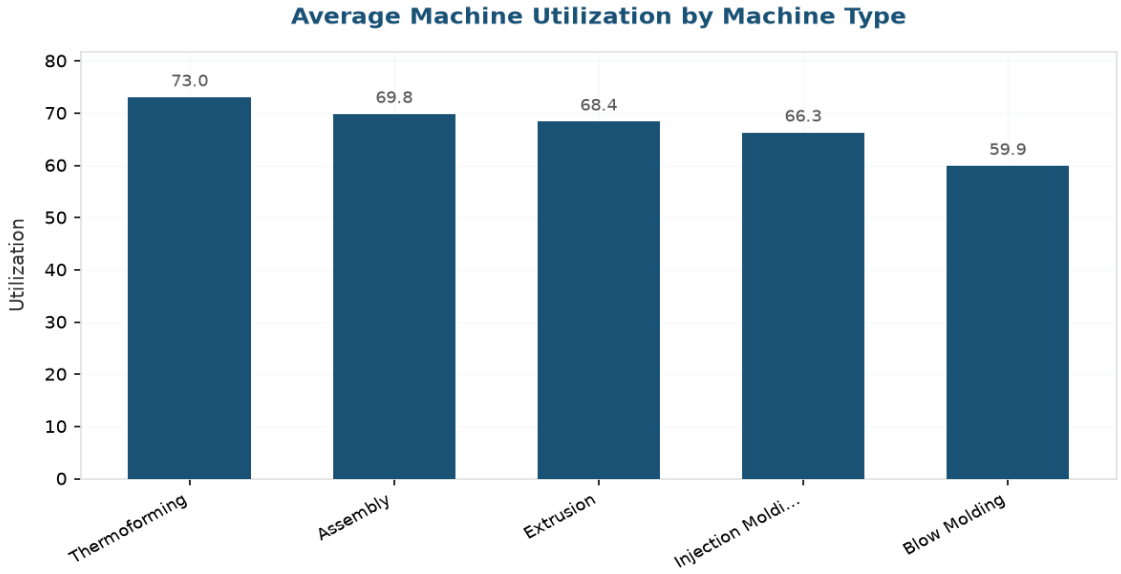
The video analytics system configures 3 distinct camera endpoints: 'Flarehub' (Camera 1), 'Luxsphere' (Camera 2), and 'Entry Gate' (Camera 6). While all 3 cameras are marked as active/online in the master configuration table, historical status logs across 50 records for camera IDs 1, 2, and 3 report an exclusively 'offline' operational status. This discrepancy between active administrative status and offline system logs points to potential telemetry disconnects or logging configuration gaps that warrant operational investigation.

Data preview — cameras (showing 3 of 3 points/rows)

camera_number	department_id	location_id	name	plant_id	status	user_id
1	None	None	Flarehub	None	online	admin
2	None	None	Luxsphere	None	online	admin
6	1	1	Entry Gate	1	online	admin

MES Machine Utilization by Machine Type

Data from the MES MachineMaster database across 40 operational machines indicates an average machine utilization rate of 68.61%, spanning from a low of 43.40% to a high of 92.70%. The fleet consists of 5 primary equipment categories: Injection Molding, Thermoforming, Assembly, Extrusion, and Blow Molding. Overall capacity averages 541.73 total units with an average hourly capacity of 216.60 units per hour, highlighting moderate overall plant efficiency.

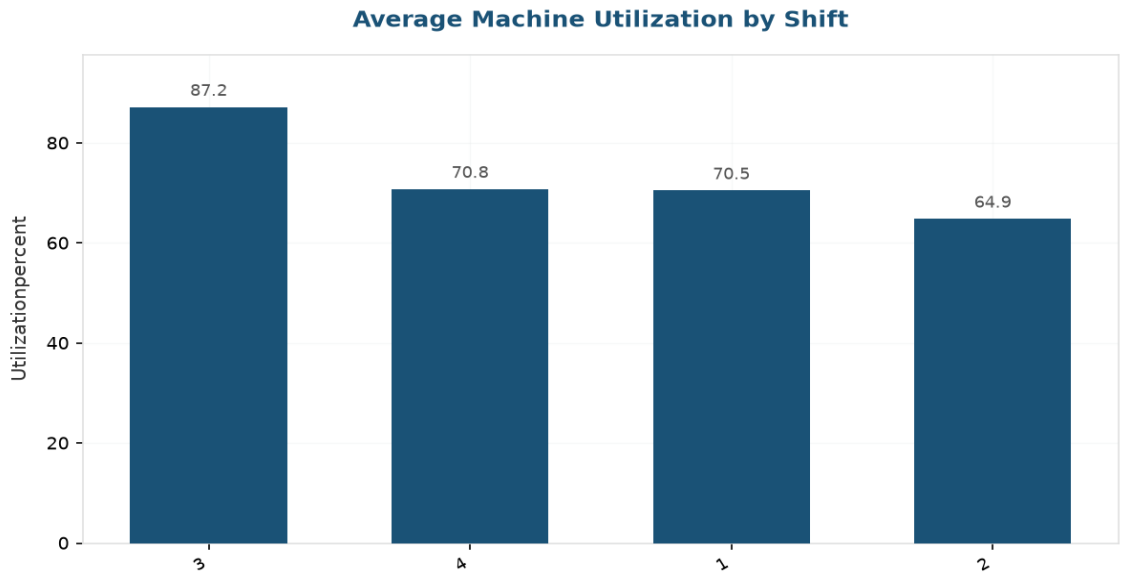


Data preview — MachineMaster (showing 10 of 40 points/rows)

Utilization	MachineCode	MachineId	MachineName	MachineType	Capacity	CapacityPerHour	CustomerId
81.90	M001	1	Injection Molding Line 1	Injection Molding	772.96	178.64	1
45.00	M002	2	Thermoforming Line 2	Thermoforming	554.94	77.34	1
58.90	M003	3	Thermoforming Line 3	Thermoforming	826.22	229.64	1
75.90	M004	4	Thermoforming Line 4	Thermoforming	681.29	150.48	1
59.00	M005	5	Assembly Line 5	Assembly	120.46	235.09	1
69.70	M006	6	Injection Molding Line 6	Injection Molding	190.81	333.32	1
43.40	M007	7	Extrusion Line 7	Extrusion	705.28	335.53	1
56.50	M008	8	Thermoforming Line 8	Thermoforming	572.02	66.41	1
77.10	M009	9	Assembly Line 9	Assembly	283.00	388.73	1
70.40	M010	10	Blow Molding Line 10	Blow Molding	943.44	217.79	1

Shift-Level Capacity Utilization Analysis

Detailed capacity logs from CapacityAnalysis evaluate machine utilization across 4 distinct shifts for key core equipment (Machine IDs 1 to 4). Average shift utilization stands at 72.92%, with individual records spanning from a minimum of 37.50% to full capacity at 100.00%. Variance across shifts highlights specific operational windows where equipment is significantly underutilized.



Data preview — CapacityAnalysis (showing 10 of 50 points/rows)

MachineId	UtilizationPercent	CapacityId	CustomerId	ShiftId
1	100.00	1	1	4
1	51.62	2	1	1
1	52.62	3	1	4
1	100.00	4	1	3
1	100.00	5	1	2
1	61.88	6	1	2
1	67.25	7	1	1
1	59.75	8	1	3
1	64.00	9	1	4
1	76.38	10	1	1

Appendix — Data Quality

Table	Records	Completeness	Outliers	Score
cameras	3	71.4%	0	Needs review
camera_status_logs	50	100.0%	0	Good
MachineMaster	40	100.0%	0	Good
CapacityAnalysis	50	100.0%	0	Good
Machine	40	100.0%	0	Good
MachineFGMapping	50	100.0%	0	Good
MachineMaterialMapping	50	100.0%	0	Good
WeighingMachine	50	100.0%	0	Good

LLM Usage & Cost

Model	Input tokens	Output tokens	Total tokens	Estimated cost
gemini/gemini-3.5-flash	5,372	2,648	8,020	$0.0319